Climate Action: Plan, Measure, Improve

GROW · Science · 40 minutes

I can explain how an action reduces climate impact, distinguish reducing causes from coping with effects, and plan evidence to check whether it worked.

Prepare before pupils arrive

Setup: Prepare three approved scenario cards and the fictional before/after data. No live site change is required.

Safety: Never interrupt safety or accessibility equipment. Do not collect family travel, food or energy data. Frame action as shared systems work.

Equipment: Scenario cards; before/after planning mat; pencil; optional local data-strip route.

Safety: Pupils do not touch sockets, plugs, boilers, thermostats or site controls. Staff approval comes before any action.

Fallback: Use the paper mat, point to cards or dictate exact words. No equipment or live site action is needed.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Describe causes and effects of global warming and how to reduce our impact.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	State how switching off an unused light could reduce emissions. Explain one difference between mitigation and adaptation.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today’s answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask ‘What makes you think that?’ and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	mitigation — reduce a cause adaptation — cope with an effect baseline — what happens before Problem: lights are sometimes left on in empty rooms, using electricity unnecessarily. Action and mechanism: switch them off safely; where fossil fuels supply energy, lower demand can mean fewer emissions.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Baseline: record what happens before the change. Prediction: state what should change and why.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Sort each card as Mitigation, Adaptation or Evidence check. Use the fictional data only to practise reasoning.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Choose one approved school scenario and create an action, mechanism and measurement plan before making any success claim. Record: A safe action-to-mechanism chain, baseline, repeated measure, comparable condition and one limitation.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Explain one action-to-mechanism chain. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Record unused light-on minutes for five days before and five days after, in the same way.

Yes. It includes a baseline, an action and repeated comparable measurements.

Misconception repair

Repaired. The evidence supports a local change in the measured minutes; the larger climate effect was not measured.

Guided checkpoint

Yes. It includes a baseline, an action and repeated comparable measurements. Look for measurements from before and after the action, taken in the same way.

Unused lights off

MITIGATION. This is mitigation because lower unnecessary energy demand can reduce greenhouse-gas emissions. Ask whether this action reduces a cause, copes with an effect or checks a result.

Safe shade in high heat

ADAPTATION. This is adaptation. It helps people cope with an effect but does not directly remove the cause. Focus on whether the action reduces emissions or reduces harm from heat.

Before: 300 minutes

EVIDENCE CHECK. This is the baseline. It records what happened before the routine began. A number from before the action belongs in the evidence check.

After: 90 minutes

EVIDENCE CHECK. Compared with the 300-minute baseline, wasted-light time fell in this fictional sample. This measurement checks what happened after the action.

A poster proves success

EVIDENCE CHECK. This is an evidence check with a weak result: displaying a poster does not show that impact changed. Test this statement against the evidence target: what was actually measured?

Model limitation

Light-on minutes suggest possible electricity savings only when the same lamps and conditions are compared; they do not measure a global climate result.

Independent evidence

A safe action-to-mechanism chain, baseline, repeated measure, comparable condition and one limitation.

Supported: Select labelled cards for Action, How it helps and What we measure. Choose between two approved scenarios and use a before/after picture strip.

Standard: Complete the full plan with a baseline, prediction, repeated measure and one controlled condition. Use the prompt: If we ___, then ___ because ___. We will measure ___.

Stretch: Compare mitigation with adaptation and identify one confounding factor or evidence limit. Explain why the result must stay local and provisional until the evidence is collected.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_G_W11B_Climate_Action_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.